

CS383 - Software Engineering

Software and Project Management

Outline

- Project management
 - Manager duties
- Project planning
 - Software pricing, project scheduling, agile planning, estimation techniques
- People management
- Risk management

Software Project Management

- Coordinating and managing tasks and software engineers involved in a project
- Successful criteria for project management are
 - Deliver the software within agreed time
 - Keep overall costs of software and its development within budget
 - Deliver software that meets requirements specification
 - Maintain motivated, happy and well-functioning development team

Software management: managers' duties

- Planning:
 - Estimating costs and scheduling project development
 - Assign people to tasks
 - Monitor development progress and it is within agreed time and budget
 - Ensure work is carried out within standards
- Reporting:
 - Report on the progress and write coherent document of critical information
- Risk management:
 - Identify risk and assess them and then take actions when problems emerge

Software management: managers (cont.d)

- People management:
 - Choose people to form teams
 - Identify ways of helping teams to lead to effective performance
- Proposal writing (to win a contract to carry out an item of work):
 - Describe objectives and how they are carried out
 - Provide cost and schedule estimates with justifications

Project planning

- Project plan includes
 - Setting intermediate goals and milestones
 - Define tasks (breaking down problems into subproblems)
 - Scheduling tasks and estimate time and resources
 - Avoid delays by minimizing task dependencies
- The project management and plan must be documented in SPM
- There are tools support that facilitate project planning
 - Jira by atlassian, Smartsheet, Microsoft Project, Microsoft Planner, Wrike, Nutcache

Project planning: ABHS example

- Consider developing software for **A**utomated **B**aggage **H**andling **S**ystems
- Define milestones
 - Requirements specifications
 - System architecture and design (including any subcomponents design)
 - Prototype 1: Baggage check-in
 - Prototype 2: Baggage routing
 - Prototype 3: Staff scheduling
 - Integrated System
- These milestones can have initial delivery fixed dates

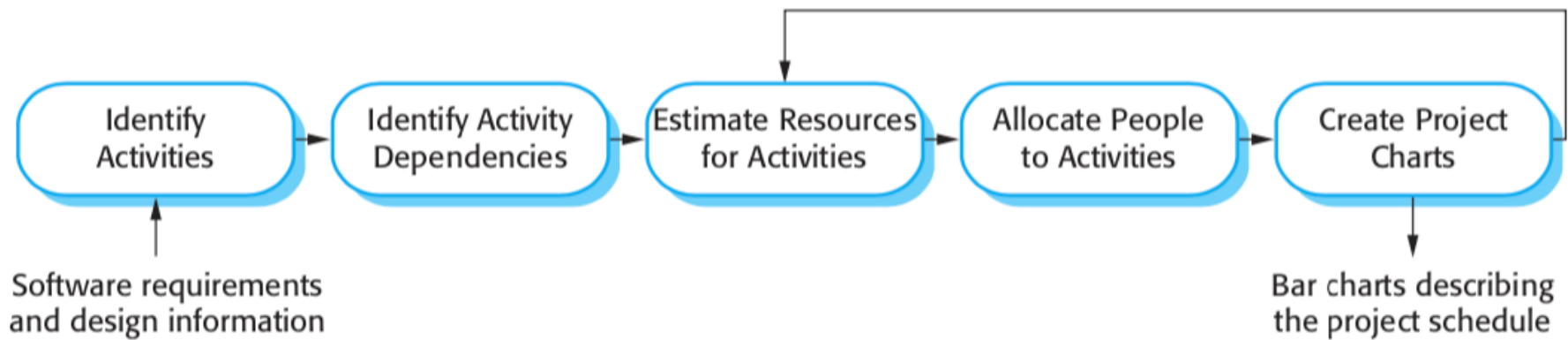


Project planning (cont.d)

- The plan, therefore, normally includes:
 - Introduction: description about the objectives and constraints (time, budget)
 - Project organization: describe the arrangement of development team and their roles
 - Risk analysis: identify risks and reduction strategies to manage them
 - Resource requirements: specifies the hardware and support software required for development
 - Work breakdown: divide the work into activities and milestones
 - Project schedule: shows the dependencies between activities, the estimate time of each milestone/deliverable, and people allocation for each activity.

Project planning (cont.d)

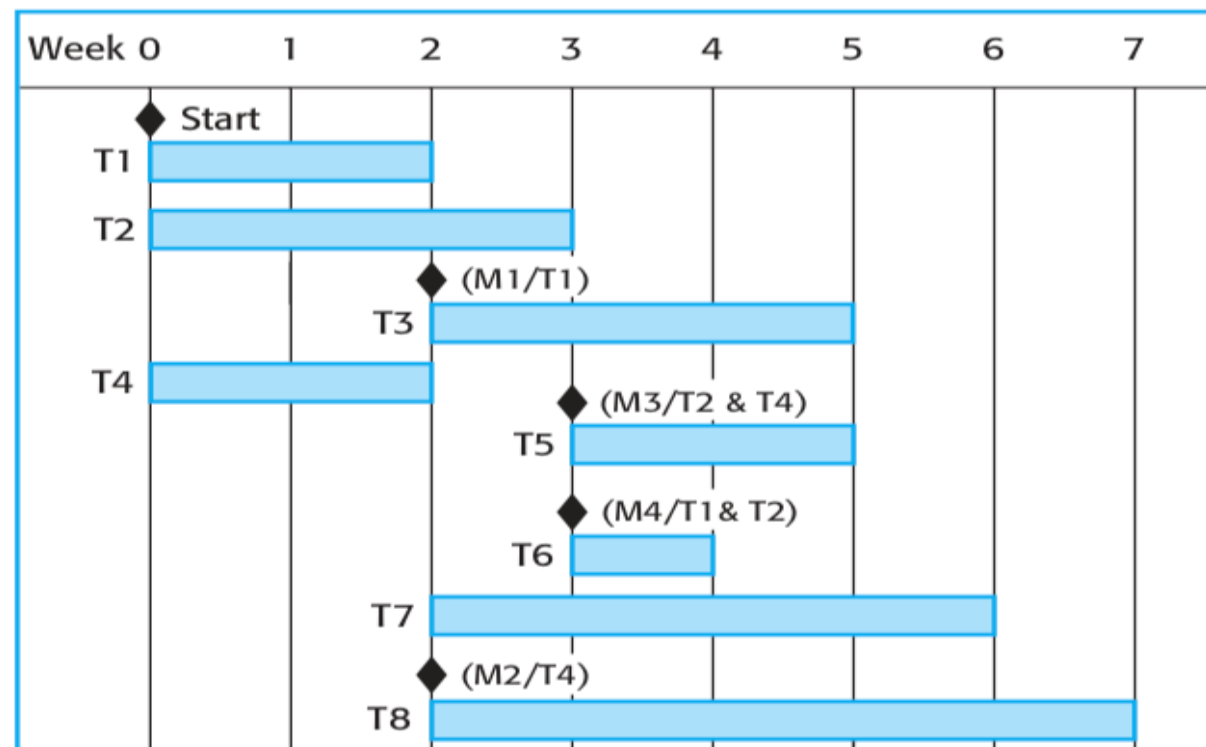
- Typical project scheduling process



- Where possible, tasks should be self-contained (why?)

Project planning (cont.d)

- Example of a project schedule chart: T means task, and M means milestone
- You can also include people involved for each task. Eg. T3 (Mohammad, Khalid, ...)



Project planning: estimation techniques

- There are two types of techniques can be used for effort estimation

1. Experience-based techniques: use manager's experience of past projects and application domains to predict the effort required.

- There are drawbacks of Experience-based techniques type:
 - Software project may not have much in common with previous project
 - If strong experience unavailable, then, calculating effort become hard and not accurate as possible.

Project planning: estimation techniques

2. Algorithmic cost modeling: formulaic approach to compute effort based on estimates of project attributes: size, process characteristic, and the staff involved (experience and skills).

- Algorithmic models for estimating effort in a software project are mostly based on a simple formula:

$$\text{Effort} = A \times \text{Size}^B \times M, \quad \text{where:}$$

- A: constant factor depends on organizational practices and type of software
- Size: either assessment of code size, functionality estimation, or application points
- B: size and complexity of the system (lies between 1 and 1.5)
- M: multiplier of combining process, product and development attributes (e.g., dependability requirements and experience of the development team)

Project planning: estimation techniques

- Nevertheless, initial estimation of any project are not 100% accurate (why?)
 - Because no empirical data is available to make the estimation
 - The actual estimation was found vary by 0.25 to 4.00 of initial estimates.
 - Eg. if initial estimation is x , then the actual estimation range from $0.25x$ to $4x$.
- Project estimation becomes more and more accurate as the project progress.

Project planning: software pricing

- Software pricing is proposing to the customer the costs of completing the project
 - Estimate of effort is requirement to complete each activity
 - Then, calculate the cost based on that and finally the price to the customer
- There are parameters that should be use when calculating the costs
 - Effort costs
 - The costs of paying software engineers and managers
 - Hardware and software costs including maintenance
 - Travel and training cost
- Software pricing can be influenced by type of organization
 - economic, political, business, education, etc

Project planning: software pricing (cont.d)

- There are also other factors affecting software pricing:
 - Market opportunity
 - Quote a low price just to move into a new segment of the software market, gain experience and to be known. Profit can be gained later.
 - Contractual terms:
 - Customer is willing to allow developing company to retain ownership of the software code, that can be used later for similar kind of applications.
 - Requirements volatility:
 - Requirements are likely to be changed, and therefore, you may propose lower price at the start. Charge later when requirements change.
 - Financial health:
 - Developers may propose low price because financial difficulty.
 - Cash flow is better than profit in financially bad times

People management

- Project manager should consider four factors in people management
 - Consistence: members should feel that their contributions are valued fairly.
 - Respect: manager should respect everyone regardless of their diverse skill-set
 - Inclusion: members should feel that others are listen to them and appreciate their contribution whatsoever
 - Honesty: manager should always be honest

People management (cont.d)

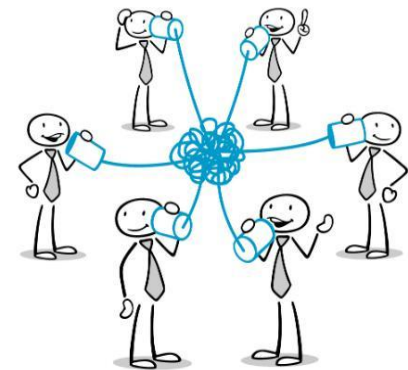
- The successful manager ensures that the group
 - Works effectively
 - Has the resources needed
 - Is well-motivated
 - Has no misunderstanding
 - Group meets its goal

People management: teamwork

- Software is developed by project teams.
- Large software projects should not be tackled by a single large team (why?)
 - Risk of spent time in communication than development
 - Managing a single large team is hard: loose tracking of tasks and their completion
- Small programming teams (less than 10 members) have a number of benefits:
 - Less communication problems
 - Easy to form cohesive team
 - Quality standard can be developed
 - Members work together closely
 - Team members get to know each others
 - Refactoring and continual improvement can be easy encouraged

Team communication

- Team members who need a regular communication
 - Status meetings: know about the project's status and progress
 - At least on weekly basis
- All means of communication are recommended
 - On-site meetings are usually very effective
 - Alternative: email, virtual meetings, telephones, formal documents ,etc.



Project management: meetings

- Group leader must organize successful meetings:
 - Meetings clear purpose and objective
 - Stick to the clear purpose: ensure no stray from objectives
 - Provide meeting agenda: list of issues to discuss
 - Time frame: meeting should starts and ends in defined times
 - Provide minutes: any decision made in the meeting should be documented

- There are two types of meeting
 - Status meetings
 - Action meetings

Project management: status meetings

- Bring participants up-to-date on other peoples portions of the project
 - A short status of report and circulate it
- Agenda should include written reports from people on the work they are responsible for.
 - New business should be kept to a minimum. your individual status report
- Status meetings are typically short: ten to fifteen minutes
 - So make sure everybody has read reports before meeting!

Project management: action meetings

1. Brainstorming session

- Craft variety of possible solutions to a particular problem
- Idea of one person often inspires ideas in other people
- A group can produce more creative ideas than a single person
- Don't throw an idea away too soon
 - They may have good properties you want to exploit or stimulate other ideas
- People have to be encouraged to put their ideas forward:
 - Never dismiss or criticize a suggestion (until a separate evaluation phase later)
 - Before starting a new idea of your own
 - restate what the previous speaker said to their satisfaction.
 - find at least one good thing to say about the previous idea.
- At the end of a brainstorming session, the ideas must be evaluated

Project management: action meetings

2. Decision-making meetings

- These meetings are fairly rare
- Decision usually falls under one person's area of authority and there is no need to agree on something
- Sometimes a decision is not clear-cut and the decision maker wants suggestions or advice from those affected by the decision
- A common type of decision-making meeting involves choosing between several alternatives
- Participants need to be familiar with the alternatives before the meeting

Project Control

- Monitor the process and activities against the plan
 - Take corrections when there is deviation from plan
 - Realistic plan is important so that corrections are minimum
 - Adding people into a behind schedule project will delay it further (why?)
- It is usual that the project plan evolves as the project progress (why?)
 - Expect that time allocations are going to adjusted
 - Very important to keep accurate records of your time spent on tasks.

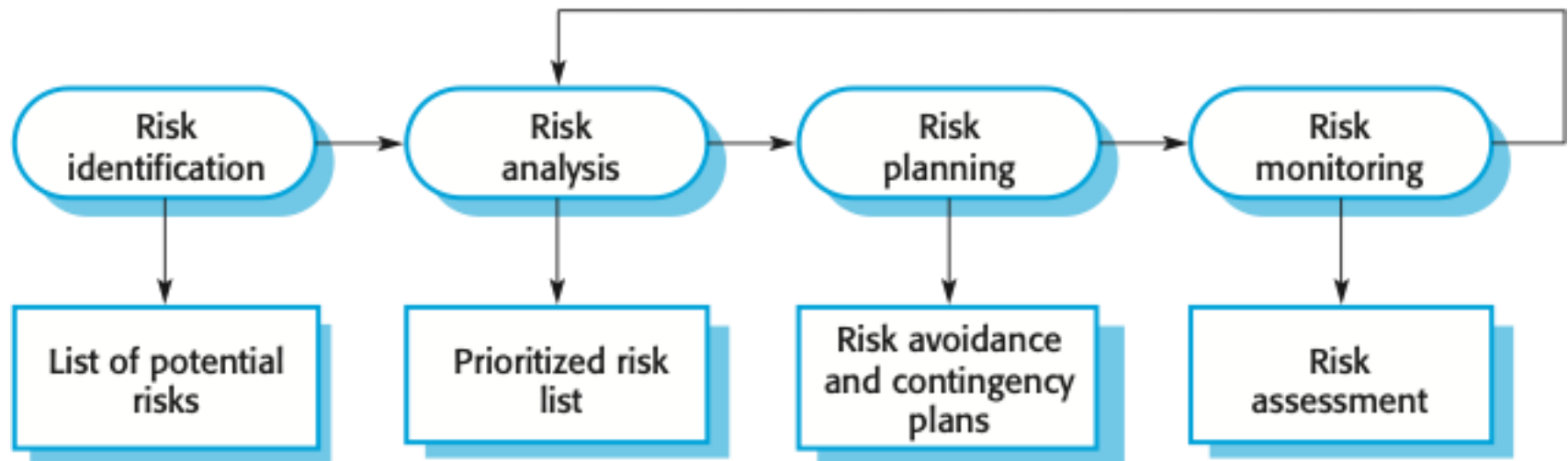


Risk management

- It involves anticipating risks that may affect the project schedule or the quality of project
- There are three categories of risks (they can overlap):
 - Project related: affect schedule or resources.
 - Eg. essential hardware for the project will not be delivered on time
 - Product related: affect the quality or performance of the product
 - Eg. the size of the system has been underestimated (affect project and product)
 - Business related: affect the organization developing the software
 - Eg. a competitive product is marketed before the current system is completed

Risk management (cont.d)

- Risk management process



Risk management: example 1

- Risk identification: organizational financial problems force reductions in the project budget.
- Risk analysis:
 - Probability: very low, **low**, moderate, high, very high
 - Impact: **catastrophic**, serious, tolerable, insignificant
- Risk planning:
 - Prepare a briefing document for senior management showing how the project is making a very important contribution to the goals of the business and presenting reasons why cuts to the project budget would not be cost-effective.

Risk management: example 2

- Risk identification: key staff is ill and unavailable at critical times in the project
- Risk analysis:
 - Probability: very low, low, **moderate**, high, very high
 - Impact: catastrophic, **serious**, tolerable, insignificant
- Risk planning:
 - Reorganize team so that there is more overlap of work and people therefore understand each other's jobs.

Risk management: example 3

- Risk identification: it is hard to recruit staff with the required skill-set for the project.
- Risk analysis:
 - Probability: very low, low, moderate, **high**, very high
 - Impact: **catastrophic**, serious, tolerable, insignificant
- Risk planning:
 - Alert customer to potential difficulties and the possibility of delays; investigate buying-in components.